

LLM-Agent-Based Energy-Neutral Random Access for Low-Altitude UAV Localization

Erchao Zhu*, Jiedan Tan*, Fang Liu, Jingwen Tong, Taotao Wang and Shengli Zhang

Abstract—In GPS-denied environments, energy-harvesting low-altitude Unmanned Aerial Vehicles (UAVs) require frequent interaction for localization assistance, facing severe challenges in energy efficiency. Existing fixed-probability or heuristic strategies struggle to balance dynamic energy arrivals with channel competition, often leading to energy depletion or restricted throughput. To address this, we propose the Reflective Localization Agent (RLA) protocol. This protocol deploys an Energy Monitor at the physical layer to ensure energy causality constraints and constructs an “Observe-Reflect-Decide-Execute” (ORDE) loop at the cognitive layer. The core mechanism leverages the semantic reasoning capabilities of Large Language Models (LLMs) to diagnose the sustainability of throughput gains (i.e., avoiding short-sighted over-consumption) and maps qualitative strategy adjustment suggestions to quantitative access probabilities. This enables nodes to spontaneously converge to an Energy Neutral operating state without the need for online training. Compared to traditional ALOHA protocols, RLA utilizes harvested energy more effectively, achieving significant throughput improvements while ensuring long-term system survival. Its performance closely approaches greedy strategies while significantly reducing outage risks. Simulation results demonstrate that RLA achieves Energy Neutrality through a cognitive closed-loop, maintaining battery levels within a robust range in varying environments.

Index Terms—Low-altitude UAVs, energy harvesting, random access protocol, agent-based communication, energy neutral operation.

I. INTRODUCTION

With the rapid development of the low-altitude economy, autonomous localization of Unmanned Aerial Vehicles (UAVs) in GPS-denied environments has become increasingly critical [1]. This process relies heavily on high-frequency communication with ground Access Points (APs) to obtain correction data [2], [3]. However, the limited capacity of onboard batteries and the stochastic fluctuations of energy harvesting environments pose severe challenges to traditional random access protocols [4], [5]. Existing access strategies based on fixed thresholds or probabilities [6] struggle to adapt to the dynamics of energy arrival: conservative strategies lead to low resource utilization, while aggressive strategies easily trigger communication outages [7], [8]. Although learning-based approaches such as Deep Reinforcement Learning (DRL) have been explored, their “black-box” nature obscures the decision-making logic and often limits adaptive transparency in highly dynamic environments.

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Recently, Large Language Models (LLMs) have demonstrated astonishing Zero-shot Planning capabilities and are regarded as a transformative paradigm for solving complex resource allocation in 6G and the low-altitude economy [9], [10]. Unlike traditional RL which requires extensive trial and error, Reinforcement Learning from Language Feedback (RL-LF) frameworks, such as Reflexion, have proven that agents can rapidly correct errors from historical interactions by introducing self-reflection mechanisms [11]. Inspired by this, our previous work has proposed an “Observe-Reflect-Decide-Execute” (ORDE) closed-loop mechanism more suitable for real-time control [12]. This mechanism effectively utilizes historical interaction information to achieve rapid adaptation to complex constrained environments without large-scale offline training.

Based on the aforementioned cognitive framework, this paper proposes a Reflective Localization Agent (RLA) protocol specifically designed for low-altitude localization and navigation scenarios. To address the energy-constrained competitive access environment, we design a lightweight reflection mechanism based on LLM semantic reasoning. RLA nodes break the reliance on preset rules typical of traditional protocols (e.g., fixed ALOHA [13]) and are capable of “thinking” like humans: they dynamically generate natural language strategy adjustment suggestions based on battery levels and historical collision feedback. These suggestions are then mapped to specific channel access parameters (e.g., transmission probability), maximizing energy efficiency while ensuring localization accuracy.

To the best of our knowledge, we are the first to introduce an LLM-driven reflection mechanism into the MAC layer for energy-constrained UAV communications. The proposed RLA protocol replaces preset access rules with semantic reasoning to provide interpretable strategy adjustments, effectively overcoming the “black-box” limitations and lack of transparency inherent in traditional learning-based methods. By integrating an ORDE closed-loop with an Energy Monitor module, our design decouples physical energy constraints from high-level access logic, enabling the agent to map numerical battery states and collision feedback into semantic descriptions for zero-shot adaptation without extensive training. Through extensive simulations, we demonstrate that RLA autonomously maintains a stable energy operating regime and significantly reduces communication outages, achieving competitive long-term throughput while ensuring survivability compared with greedy and heuristic-based ALOHA strategies.

The remainder of this paper is organized as follows: Sec-

tion II introduces the system model and problem formulation; Section III describes the RLA protocol design; Section IV presents the experimental results and analysis; and Section V concludes the paper.

II. SYSTEM MODEL

A. Network Scenario

This paper focuses on low-altitude localization and navigation scenarios in GPS-denied environments [2], [14]. We consider a random access wireless network composed of M Unmanned Aerial Vehicles (UAVs) and an Access Point (AP).

Time is discretized into fixed-length slots. We adopt a saturated traffic model, where each UAV has a fresh status update available at the beginning of every slot. Transmissions are synchronized and can only be initiated at slot boundaries, with each transmission completed within a single slot. If multiple UAVs transmit simultaneously, a collision occurs, resulting in unsuccessful transmissions. At the end of each slot, the AP broadcasts a lightweight feedback message indicating the channel access outcome (idle, success, or collision), which can be exploited by UAVs for adaptive access decisions.

B. Energy Harvesting and Battery Dynamics

Due to the substantial power consumption required for flight propulsion, the energy budget available for onboard communications is severely limited. As a result, the communication module must rely on ambient energy harvesting to sustain its operation [4]. Each UAV is equipped with a rechargeable battery with a finite capacity of $B_{\max}$ energy units.

Following [6], [7], the energy harvesting process is modeled as an independent Bernoulli random process $H_t \in \{0, 1\}$. Specifically, in each time slot, one unit of energy is harvested with probability p_h , i.e., $\Pr(H_t = 1) = p_h$, and no energy is harvested otherwise. When the battery is full, newly harvested energy is discarded. The energy consumption for a single transmission is normalized to $E_{tx} = 1$. Let $B_t \in \{0, \dots, B_{\max}\}$ denote the battery state at the beginning of slot t . The battery evolution is given by

$$B_{t+1} = \min\{B_t - \alpha_t E_{tx} + H_t, B_{\max}\}, \quad (1)$$

where $\alpha_t \in \{0, 1\}$ is the transmission decision indicator. The action must satisfy the energy causality constraint $B_t \geq \alpha_t E_{tx}$, ensuring that transmissions can only occur when sufficient energy is available.

C. Optimization Objective

In the considered GPS-denied localization scenario, long-term average throughput is used as a tractable surrogate for localization service availability. The objective of this paper is to maximize the long-term average throughput $\mathcal{T}$ of the intelligent UAV under energy harvesting constraints. Specifically, the long-term average throughput is defined as

$$\mathcal{T} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[S_t], \quad (2)$$

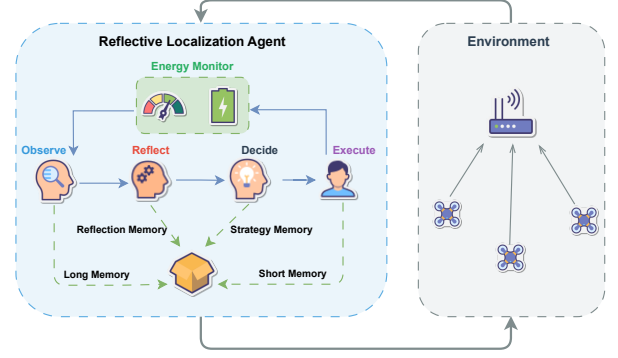


Fig. 1. RLA protocol architecture and its cognitive decision loop. This architecture introduces an Energy Monitor module, coupled with the ORDE reasoning cycle to achieve energy-efficiency-driven random access control.

where $S_t = 1$ indicates a successful transmission in slot t , and the expectation is taken over the stochastic channel access outcomes and energy harvesting process.

To this end, we propose the RLA protocol, which aims to maximize $\mathcal{T}$ through adaptive reflection and access decisions, while satisfying the energy causality constraint $B_t \geq \alpha_t E_{tx}$ that ensures energy neutral operation.

III. RLA PROTOCOL DESIGN

Here we present the proposed RLA protocol, which is based on the Reflex-Core framework devised in [12]. The protocol aims to achieve a dynamic balance between throughput and sustainability in energy-constrained environments through the interaction of an ORDE closed-loop mechanism and an Energy Monitor. To avoid redundancy, this section focuses on the architectural abstraction and decision logic of the RLA protocol. Detailed algorithm instantiation and foundational inference processes for Reflex-Core can be found in [12]. However, as the original framework does not account for energy dynamics, the energy-harvesting constraints and power-sustainability mechanisms introduced here represent novel extensions specifically designed for the RLA protocol.

A. Architectural Components and Functional Division

As shown in Fig. 1, the RLA protocol consists of three complementary layers:

- **Energy Monitor (Physical Sensing and Constraint Layer):** Serving as the physical layer “gatekeeper”, this module monitors the battery level B_t in real-time and enforces the energy causality constraint. If $B_t < E_{tx}$, the transmission action is intercepted. It abstracts raw energy dynamics into semantic states (e.g., “Surplus,” “Depletion Warning”), providing a basis for high-level reasoning.
- **Memory Module (Support Layer):** Utilizes hierarchical storage: Short memory caches slot-level trajectories and battery fluctuations; Long memory records energy efficiency trends and harvesting patterns; Reflection memory

archives historical linguistic insights and cognitive evolutions to track the agent’s reasoning progress; and Strategy memory preserves historically preferred solutions.

- **ORDE Cognitive Decision Loop (Intelligent Reasoning Layer):** Acting as the functional nucleus of the Reflex-Core, this layer integrates four specialized modules—Observe, Reflect, Decide, and Execute—to operate within the physical boundaries defined by the Energy Monitor. Through the synergy of numerical perception and semantic reflection, the loop achieves an optimal trade-off between throughput and energy sustainability.

B. Closed-Loop Evolution Logic Based on Energy-Efficiency Trade-offs

To harmonize high-frequency network dynamics with low-frequency LLM reasoning, RLA employs a hierarchical temporal structure consisting of three scales: *Fine-grained Execution* (slot level), *Aggregated Observation* (period N), and *Asynchronous Reflection* (cycle O). Specifically, an observation period aggregates metrics across N time slots to filter stochastic noise, while a reflection cycle triggers the LLM’s cognitive optimization every O observation periods. This hierarchy ensures computational efficiency while maintaining real-time responsiveness to energy fluctuations. The evolution logic for each module is as follows:

- 1) **Observe:** The module aggregates data within a window of period N , calculating the average battery level and the energy harvesting-consumption gradient through a sliding window. Based on the deviation between the current battery level and the implicit safety energy level, it generates an observation perturbation term Δp_i for node i , achieving instantaneous semantic mapping of the current “energy pressure.”
- 2) **Reflect:** The core cognitive phase is triggered every O observation periods. The LLM, combined with long-term memory, diagnoses whether the throughput gain $\Delta \tau_i$ is achieved at the cost of unsustainable battery loss. By analyzing ineffective energy consumption caused by collisions, it generates qualitative modification suggestions to suppress overly aggressive access behaviors that lead to excessive collision-induced energy waste and long-term battery depletion.
- 3) **Decide:** Based on semantic reflection suggestions, the transmission probability p_i is dynamically updated. This update is manifested as a qualitative adjustment of the probability, aiming to guide the battery level toward a robust equilibrium range, maintaining high throughput while preserving resilience to stochastic energy arrivals using reserved energy.
- 4) **Execute:** The module implements the transmission with the final transmission probability p_i^{final} , which is obtained by combining the probability p_i with the local perturbation Δp_i . Before action execution, real-time battery verification must pass through the Energy Monitor to strictly satisfy energy causality constraints. Final transmission feedback and energy consumption data are

immediately stored in the memory module to drive the next iteration.

Overall, RLA implicitly balances throughput performance and energy sustainability through its cognitive closed-loop, rather than introducing explicit control constraints. By analyzing the long-term trends of energy harvesting, consumption, and collision-induced waste in the Reflect stage, RLA suppresses unsustainable aggressive behaviors and maintains a stable operating regime across different energy harvesting rates p_h .

Due to space constraints, we show the prompts for the Reflect module serving as a representative in the Appendix and omit the prompts for other modules.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

We conduct simulations to evaluate the performance of RLA in optimizing throughput while satisfying energy harvesting constraints within a heterogeneous random access network. We consider a network with $M = 4$ UAVs, consisting of one RLA-enabled intelligent node and three nodes employing the conventional ALOHA protocol. In this competitive environment, energy arrival follows a Bernoulli distribution with $p_h \in \{0.1, 0.15, 0.2, 0.25\}$, and the battery capacity is capped at 12 units. For benchmarking, the ALOHA-based nodes are categorized into two types: one *Greedy ALOHA* node ($p = 0.8$) and two *Conservative ALOHA* nodes ($p = 0.08$), representing aggressive throughput seeking and long-term energy hoarding, respectively. RLA parameters are set to $N = 200$ slots and $O = 3$ periods. Reasoning tasks are offloaded to an edge server using DeepSeek-V3, consistent with the AI-native 6G collaborative paradigm.

As illustrated in Fig. 2 and Fig. 3, the simulation results demonstrate that RLA effectively achieves energy-neutral operation while maximizing communication performance. Regarding throughput, RLA shows strong competitiveness across all p_h settings; specifically, at $p_h = 0.25$, RLA reaches a throughput of 0.161, which is merely 1.8% lower than the 0.164 achieved by Greedy ALOHA. This indicates that RLA’s performance closely approaches greedy strategies while significantly reducing outage risks by maintaining a sustainable battery level. In contrast, while Conservative ALOHA suffers from low throughput (0.042) due to its excessive energy hoarding, Greedy ALOHA frequently triggers battery depletion and subsequent system outages despite its higher instantaneous performance. Overall, RLA demonstrates a spontaneous capacity to balance throughput gains with long-term survivability, effectively avoiding the catastrophic energy exhaustion seen in legacy heuristic protocols.

Behavioral Analysis: To further investigate the underlying mechanism, we provide a detailed behavioral analysis below. It is observed that RLA’s performance closely approaches greedy strategies while significantly reducing outage risks. This phenomenon induces a dynamic, self-organized balance between performance maximization and outage risk mitigation. Rather than converging to a static, high-risk value,

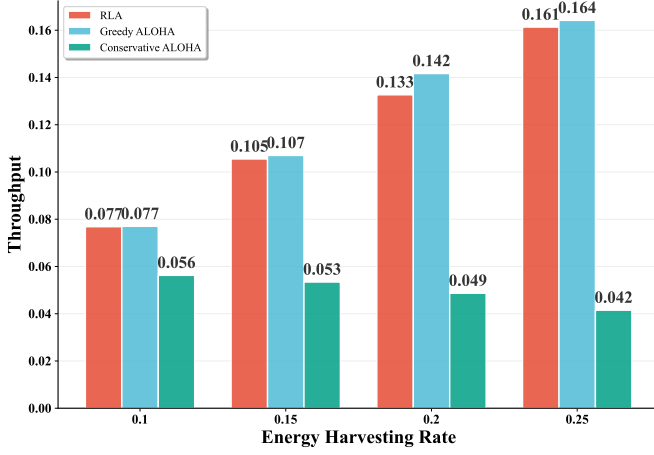


Fig. 2. Throughput comparison across different energy harvesting rates.

the protocol demonstrates a resilient adaptation to environmental uncertainty, ensuring the node remains operational even under fluctuating energy supply. This spontaneous regulation underscores the advantage of LLM-based reasoning in navigating the non-linear trade-offs inherent in energy-constrained communications. Specifically, RLA's precision in battery control manifests the effective linkage between the Energy Monitor and the ORDE cognitive loop. When p_h increases, the Observe module captures the positive energy gradient, prompting the Reflect module to relax the degree of conservatism and increase p_i for higher throughput. Conversely, when the battery level approaches a critical depletion zone, the reflection mechanism proactively identifies the risk of system failure and suppresses access intensity before energy exhaustion occurs. Such adaptive characteristics explain the resilient battery evolution and competitive throughput, proving RLA's robustness in navigating the dynamic trade-offs of energy-constrained low-altitude scenarios.

V. CONCLUSION

This paper proposes an energy-aware RLA protocol for UAV random access under energy harvesting constraints. By decoupling physical energy causality from high-level semantic reasoning via an Energy Monitor, RLA adaptively adjusts access strategies in a zero-shot manner without online training. Simulation results demonstrate that RLA maintains a stable battery operating regime and achieves throughput comparable to greedy strategies, while significantly reducing outage risks. These results highlight the potential of lightweight semantic reasoning for handling energy-efficiency trade-offs in energy harvesting-enabled UAV networks.

Future work will focus on evaluating the signaling overhead and latency of the cloud-edge collaborative architecture. Additionally, we plan to investigate the scalability of the RLA protocol in dense UAV swarms and explore the potential of using distilled small language models (SLMs) for local on-board inference.

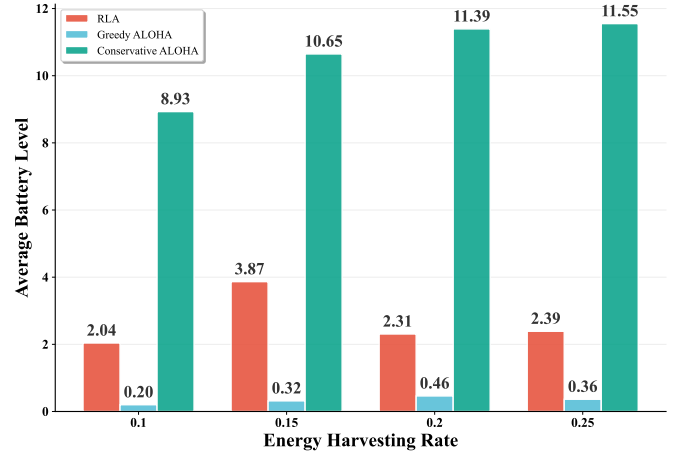


Fig. 3. Average battery level comparison across different energy harvesting rates.

APPENDIX

Reflect Module Prompt

Prompt: You are the Reflect Module of the RLA protocol. Your task is to evaluate the energy-efficiency performance of the current cycle and optimize the baseline transmission probability p_i . Focus on diagnosing whether the observation perturbation sequence Δp_i generated in this cycle was sufficient to compensate for the energy pressure brought by stochastic harvesting fluctuations.

Input Information:

1. Long-term Memory: Execution sequence of battery fluctuations and observation perturbations Δp_i for this cycle $\{\text{long_term_memory}\}$
2. Global Strategy: Baseline transmission probability set in the previous cycle $p_i = \{\text{strategy_output}\}$
3. Energy State Assessment: Average battery change in the previous cycle $\Delta B_{avg} = \{\text{energy_delta_avg}\}$
4. Historical Feedback: Diagnostic conclusions and suggestions from the last reflection $\{\text{last_reflection}\}$

Task Requirements:

1. Strategy Energy-Efficiency Evaluation: Analyze the adjustment trends of the perturbation sequence Δp_i and the battery level. Determine if the current global configuration has led to an unsustainable negative energy balance.
2. Improvement Plan Design: Propose a numerical recommendation for the probability p_i in the next cycle to ensure the system satisfies energy neutrality constraints, and guide the node's battery level back to a safe and sustainable operating range to avoid long-term energy depletion or overly conservative operation.

Example Reasoning Output:

1. *Strategy Energy-Efficiency Evaluation:* The average battery change in this cycle is $\Delta B_{avg} = -2.1$, resulting in the average remaining battery falling to 1.3 units, which is significantly lower than the energy level required for stable long-term system operation.
2. *Improvement Plan Design:* It is recommended to adjust the transmission probability p_i downward to 0.35. Current local fine-tuning can no longer correct the energy deviation; the system must prioritize rebuilding energy reserves by reducing the baseline power consumption duty cycle to ensure long-term sustainability in stochastic power supply environments.

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